

Understanding Logs & Log Analysis Basics

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1. Objective

The goal of this task was to build a foundation in log analysis: understanding the different types of logs and where they come from, learning to read a web-server log line field by field, distinguishing normal from suspicious log entries, and understanding how a security analyst uses a “baseline” of normal activity to spot anomalies.

2. Five Key Log Types

The following are five common log types an analyst works with, and the source that generates each one.

Log Type	Where it comes from
System log	Operating system events — startup/shutdown, driver errors, hardware/service failures.
Security log	Authentication and audit events — logons, logoffs, permission and policy changes.
Application log	Software/application-level events and errors generated by installed programs.
Network log	Firewalls, routers, and switches — records of traffic allowed, blocked, or routed.
Web server log	Web servers (Apache/Nginx/IIS) — records every visitor request to a website.

3. Breaking Down an Apache Web-Server Log Line

A sample Apache access log line was analysed field by field:

192.168.1.10 - - [30/Aug/2026:14:23:11 +0500] "GET /login.php HTTP/1.1" 200 3456

Part	Value	Meaning
IP Address	192.168.1.10	The client/computer that sent the request.
Identity / User	- -	Ident and authenticated username fields (usually unused, shown as ‘-’).
Timestamp	[30/Aug/2026:14:23:11 +0500]	Date, time, and timezone the request was received.
Request	“GET /login.php HTTP/1.1”	HTTP method (GET), the requested path (/login.php), and protocol version.
Status Code	200	Server's response code — 200 means the request succeeded.
Size	3456	Size of the response returned, in bytes.

General format: IP - - [timestamp] "METHOD /path HTTP/version" status_code size

Common status codes: 200 = OK, 403 = Forbidden, 404 = Not Found, 500 = Internal Server Error.

4. Normal vs Suspicious Log Entry

Normal entry

```
192.168.1.10 - - [30/Aug/2026:14:23:11 +0500] "GET /home.php HTTP/1.1" 200 3456
```

A single user accessing the homepage during typical daytime hours, with a successful (200) response — ordinary, expected activity.

Suspicious entry

```
45.67.89.12 - - [30/Aug/2026:03:14:02 +0500] "GET /admin.php HTTP/1.1" 403 512
```

```
45.67.89.12 - - [30/Aug/2026:03:14:05 +0500] "GET /wp-login.php HTTP/1.1" 403 512
```

```
45.67.89.12 - - [30/Aug/2026:03:14:08 +0500] "GET /config.php HTTP/1.1" 403 512
```

This pattern is suspicious for several reasons:

- Occurs at 3 AM — an unusual time for legitimate user activity.
- The same IP address repeatedly targets sensitive paths (admin, login, config).
- Every attempt returns 403 (Forbidden), indicating repeated unauthorized access attempts — a possible reconnaissance or attack pattern.

5. Understanding a “Baseline”

A baseline is the normal pattern of activity an analyst establishes by observing a system over time — for example, typical traffic during office hours, or the usual number of requests a single user makes per day. Once a baseline is known, any activity that deviates from it — such as a sudden spike in requests, or login attempts in the middle of the night — stands out as an anomaly and is flagged for further investigation. Baselineing is what allows an analyst to tell normal behaviour apart from suspicious behaviour, as illustrated in Section 4 above.

6. TryHackMe Rooms Completed (Mandatory Free-Tier Proof)

Both required TryHackMe rooms were completed on the free tier, as mandated by the task.

6.1 Intro to Logs

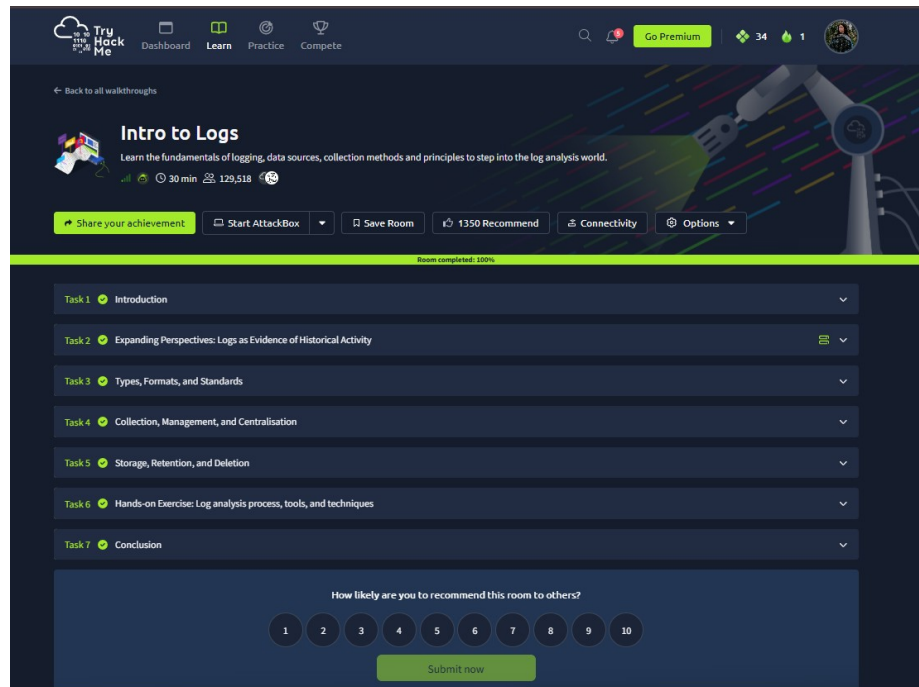


Figure 1: “Intro to Logs” room — Room completed: 100%.

6.2 Investigating Windows

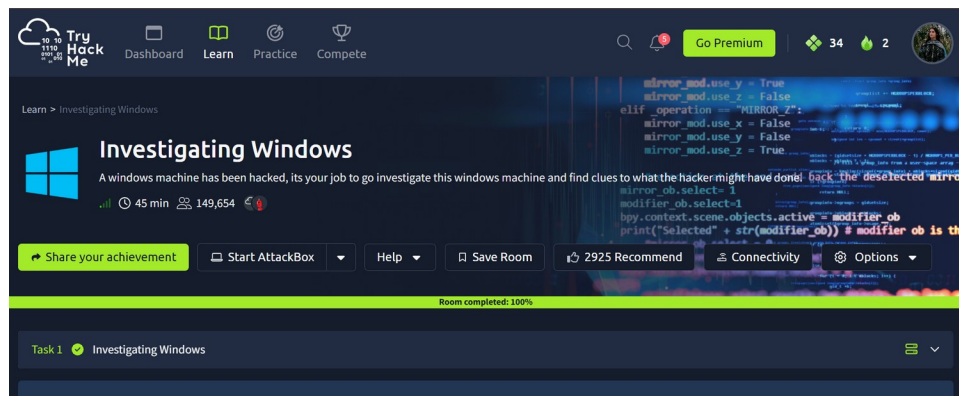


Figure 2: “Investigating Windows” room — Room completed: 100%.

7. Conclusion

This task built the conceptual foundation needed for practical log analysis: recognising the different log sources on a system, reading and interpreting a raw web-server log line, and applying the idea of a baseline to separate normal activity from suspicious activity — a core analytical skill for a SOC Analyst.